

Techno-economic and environmental assessment of CO₂ conversion to methanol: Direct versus indirect conversion routes

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ABSTRACT

Methanol as a clean energy source is one of the most important industrial petrochemical products that it is used as a fuel, solvent, and intermediate substance for other component production. Methanol is directly synthesized from CO₂ in the direct process, while in the indirect one, CO₂ is converted to CO in a water gas shift reactor as an intermediate. In this research, both direct and indirect processes are designed and simulated via Aspen Plus simulator at steady state condition for comparing not only their efficiencies, but also their economic and environmental analysis for the first time. The optimum operating conditions of processes are obtained to achieve the maximum methanol production at a fixed feed rate of 228,417 tons/year CO₂ waste gas supply. Furthermore, the best process is selected by TOPSIS decision making method by comparing the capital cost and environment issues of the direct and indirect processes. The results show that direct methanol synthesis has a superior environmental and financial efficiency compared to the indirect process as the fixed capital investment is 506 million USD for the direct process and 578 million USD for the indirect method. This is while the net profit amount is opposite, 132 million USD for direct and 135 million USD for indirect processes with the payback periods of about 7 and 8 years for the direct and indirect processes, respectively.

1. Introduction

In recent decades, the greenhouse gasses emissions by human activities have been increased significantly. The current tendency towards the defossilization of chemical industry and the energy sector is calling for the development of new processes, utilizing unconventional and renewable resources [1–3]. One of these is the development of efficient processes to capture and recycle CO₂ from effluent flue gasses from industrial processes and conversion to valuable product [1,4,5]. Over the last few years, the methanol production from CO₂ became the focus of many researchers [3]. The hydrogenation of CO₂ to methanol can reduce CO₂ emission and establish a sustainable carbon circuit [2,3]. Methanol as a clean energy source is one of the most important industrial petrochemical products that it is used as a fuel, solvent, and intermediate substance for other component production [6].

Generally, the first commercial methanol plant was designed based on the wood distillation in 1830. Due to limitations of wood distillation technology, ICI commercialized the high-pressure methanol synthesis from syngas on ZnO–Cr₂O₃ catalyst [7]. The development of superior Cu/ZnO/Al₂O₃ catalyst could moderate the operating conditions in the

methanol synthesis plant and, the low-pressure syngas conversion to methanol was commercialized by ICI at 1980 [8]. Dincer and Acer reviewed the syngas production routes from environmental impacts, investment and operating costs, and efficiency viewpoints [9]. Although methanol is commercially produced from mixture of hydrogen and carbon monoxide, the CO₂ conversion to methanol is a promising environmental-friendly technology [4]. This method is a useful strategy of CO₂ utilization and a practical approach to sustainable development [10,11]. In general, reverse water gas shift reaction and direct hydrogenation are two common routes to produce methanol from CO₂ rich feedstock. The methanol production from direct CO₂ hydrogenation results in significantly less byproducts and requires less energy in product purification compared to the conventional process [12]. However, the methanol production cost via direct CO₂ hydrogenation is 2–2.5 times higher than the cost of the conventional process and consumes more utilities than conventional methanol production [13]. In the reverse water gas shift route, the shift and the methanol synthesis reactors are serially aligned to form methanol from CO₂. Pilot and demonstration plants based on CO₂ hydrogenation using renewable hydrogen were realized by Mitsui Chemicals Company in Japan with a capacity of 100 ton per year, and by Carbon Recycling International and

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Nomenclature

Abbreviation

CAMERE	Carbon dioxide hydrogenation to methanol via reverse water gas shift
CR	Conventional route
MeOH	Methanol
RWGS	Reverse water gas shift
NCR	Net carbon dioxide reduction
CRF	Capital recovery factor
C	Concentration of component (mol m ⁻³)
d _p	Particle diameter (m)
E	Activation energy for reaction (kJ kmol ⁻¹)
f _i	Fugacity of component (Pa)
K	Adsorption equilibrium constant
R	Gas constant (kJ kmol ⁻¹ K ⁻¹)
r _i	Rate of reaction for ith reaction (mol kgcat ⁻¹ s ⁻¹)
T	Temperature (K)
f _M	Correction factor for materials of construction
f _p	Correction factor for design pressure
f _T	Correction factor for design temperature

Greek letter

η	Effectiveness factor
ΔH	Heat of reaction (kJ kmol ⁻¹)

operating in Iceland with a capacity of 4000 ton per year [8,14]. Such technical and financial efforts, although developed under peculiar circumstances characterizing the general context, show that a concrete economic interest accompanies research developments in this field. These can be considered "green processes", especially when also H₂ is obtained by green and renewable sources such as the electrolysis of water [8,15]. Van-Dal et al. [16] proposed a process to produce methanol from CO₂ via the hydrogenation route. The required CO₂ was supplied from the flue gasses of a coal power plant while water electrolysis was the hydrogen source. Bellotti et al. [17] studied methanol production through hydrogenation of captured CO₂ from a power plant by hydrogen obtained by a PEM electrolyzer. An economic feasibility study was carried out considering three different plant sizes and a sensitivity analysis was performed varying the methanol selling price, the oxygen selling option for industrial applications, and the capital cost of PEM electrolyzers. This analysis showed that the PEM electrolyzer affects the total capital investment cost significantly, while a larger plant leads to even higher capital investment. Anicic et al. [18] provided a comparison between two direct and indirect CO₂ conversion to methanol technologies in terms of economical and energy-efficiency efficiencies. It was shown that the direct CO₂ conversion process had higher energy and economic efficiency. Loenzio G. et al. [8] investigated the production of methanol from the hydrogenation of carbon dioxide in a direct method, in which three configurations were used; without recycle flow, with recycle stream, and in the presence of the membrane. According to this study, the reactor with return flow including unreacted gasses has the best performance and the highest conversion rate in the operating conditions of 200 °C and 55 bar. Borisut P. et al. [19] optimized hydrogenation of carbon dioxide for methanol production via response surface methodology (RSM) at inlet pressure to the first reactor: 57.8 bar, Inlet temperature to the first reactor: 183.6 °C, Inlet pressure to the second reactor: 102.6 bar. They achieved the lowest level of the economic costs per ton of methanol production 565 \$. Yusuf N. et al. [20] simulated the process of methanol production from carbon dioxide hydrogenation in an adiabatic reactor operated with a CO₂/H₂ ratio of 1:7, temperature of 210 °C and pressure of 70 bar using Aspen Plus software. The obtained results depicted the production of methanol

with more than 99.84% yield and a CO₂ conversion of 95.66%. The economic assessment showed the payback period of about 9 years for the proposed process.

The main aim of this study is to present an economic and eco-friendly process to produce methanol from flue gas as a CO₂ source. In this regard, two direct and indirect CO₂ conversion processes are designed and the technical, environmental, and economic feasibility of the proposed processes are compared. In the direct process, the CO₂ is directly hydrogenated over the heterogenous catalyst, while CO₂ is reformed to CO and produced syngas is converted to methanol in a different unit in the indirect process. To compare the performance of the proposed process, the flowsheets are simulated by the Aspen Plus simulator.

2. Process description

Fig. 1 shows the generic processing scheme of methanol synthesis from CO₂ and H₂. Basically, the reactants are heated and pressurized to the desired conditions and fed into a reactor. In the direct route methanol is produced in the reactor, while CO is produced as an intermediate in the indirect route. To separate the condensable compounds from the non-condensable, the outlet stream from the reaction section is cooled and fed into a two-phase separator. The unreacted hydrogen, CO₂ are recycled to the methanol synthesis reactor. The liquid stream fed to the distillation section to separate methanol and water.

3. Process simulation

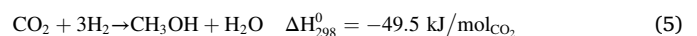
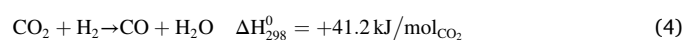
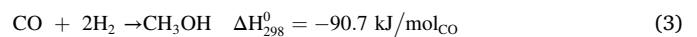
The proposed processes are simulated in the Aspen Plus simulator software at steady-state conditions. Aspen Plus software has been supported by a comprehensive database and a detailed and accurate thermodynamic package. In this research, since the Redlich-Kwong equation of state could predict the properties of components, it is chosen as the fluid package. In the indirect method, the adiabatic tabular reactor is used for the shift reactor. The considered reverse water gas shift reaction is as:



The kinetic of reverse water gas shift reaction is described as below [21]

$$r = k_o e^{-\frac{E}{RT}} \left[C_{\text{CO}_2} - \frac{C_{\text{CO}} C_{\text{H}_2\text{O}}}{K_c C_{\text{H}_2}} \right] \quad (2)$$

In both processes, the water-cooled tabular reactor is used for the CO₂ hydrogenation reactor. The considered reactions in the CO₂ hydrogenation reactor are [22]:



The developed model by Graaf et al. is used in the kinetic model section [21,23]:

$$r_1 = \frac{k_1 K_{\text{CO}} \left[f_{\text{CO}} f_{\text{H}_2}^{3/2} - \frac{f_{\text{CH}_3\text{OH}}}{f_{\text{H}_2}^{1/2} K_{p1}} \right]}{(1 + K_{\text{CO}} f_{\text{CO}} + K_{\text{CO}_2} f_{\text{CO}_2}) \left[f_{\text{H}_2}^{1/2} + \left(\frac{K_{\text{H}_2\text{O}}}{K_{p2}^{1/2}} \right) f_{\text{H}_2\text{O}} \right]} \quad (6)$$

$$r_2 = \frac{k_2 K_{\text{CO}_2} \left[f_{\text{CO}_2} f_{\text{H}_2} - \frac{f_{\text{H}_2\text{O}} f_{\text{CO}}}{K_{p2}} \right]}{(1 + K_{\text{CO}} f_{\text{CO}} + K_{\text{CO}_2} f_{\text{CO}_2}) \left[f_{\text{H}_2}^{1/2} + \left(\frac{K_{\text{H}_2\text{O}}}{K_{p2}^{1/2}} \right) f_{\text{H}_2\text{O}} \right]} \quad (7)$$

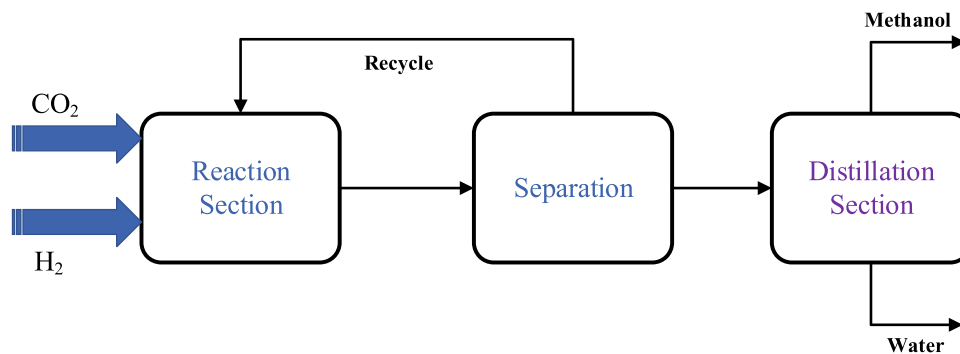


Fig. 1. Generic processing scheme of methanol synthesis from CO₂ hydrogenation.

$$r_3 = \frac{k_3 K_{CO_2} \left[f_{CO_2} f_{H_2}^{3/2} - \frac{f_{CH_3OH} f_{H_2O}}{f_{H_2}^{3/2} K_{p3}} \right]}{(1 + K_{CO_2} f_{CO_2} + K_{CO_2} f_{CO_2}) \left[f_{H_2}^{1/2} + \left(\frac{K_{H_2O}}{K_{H_2}^{1/2}} \right) f_{H_2O} \right]} \quad (8)$$

4. Process assessment

4.1. Process analysis

4.1.1. Sensitivity analysis

The sensitivity analysis investigates the effect of changes in input variables on target variables. Also, using sensitivity analysis, the value of a design or operational variable can be determined in such a way that specifies the target variables. In this research, the effect of feed flow rate and operating variables including pressure, temperature, and composition on the methanol production rate and yield is investigated by the developed simulator.

4.1.2. Economic analysis

The fixed and production cost of proposed processes are evaluated according to the economic analysis proposed by Smith and Peters [24, 25]. Initially, sufficient capital must be committed to constructing all aspects of the facility necessary for the plant. Since net profit equals total income minus all expenses, the chemical engineer must be aware of the various types of costs associated with each manufacturing step. Funds must be available for direct plant expenses, such as those for raw materials, labor, and utilities, and indirect expenses, such as administrative salaries, product sales, and distribution costs. In this section, investment and plant operation costs are reviewed as well as cash flow and gross and net profits [24]. The capital investment cost of the methanol synthesis reactor and distillation column is obtained through:

$$C_E = C_B \left(\frac{Q}{Q_B} \right)^M \quad (9)$$

where C_E and C_B are the equipment cost with desired capacity Q and base capacity Q_B , and M is the constant depending on equipment type. The costs are further corrected with the Chemical Engineering's Plant Cost Indexes [26]. Also, the cost of compressors, flash drums, and heat exchangers could be estimated as [24,25]:

$$C_{Com} = 9.84 \times 10^4 \left(\frac{Q}{250} \right)^{0.46} \left(\frac{607.5}{435.8} \right) f_M f_P f_T \quad (10)$$

$$C_{Heater} = 3.28 \times 10^4 \left(\frac{Q}{80} \right)^{0.68} \left(\frac{607.5}{435.8} \right) f_M f_P f_T \quad (11)$$

$$C_{Drum} = 1.5985 \times 10^4 \left(\frac{Q}{1.5} \right)^{0.49} \left(\frac{607.5}{435.8} \right) f_M f_P f_T \quad (12)$$

The cost of the empty distillation column and sieve tray could be

estimated as [24,25]:

$$C_E = 6.56 \times 10^4 \left(\frac{Q}{8} \right)^{0.89} \left(\frac{607.5}{435.8} \right) f_M f_P f_T \quad (13)$$

$$C_E = 6.56 \times 10^3 \left(\frac{Q}{1} \right)^{1.2} \left(\frac{607.5}{435.8} \right) f_M f_P f_T \quad (14)$$

The operating costs consist of the cost of raw materials, electricity, maintenance, labor, etc. [27]. Other expenses, including costs of the cooling water and catalyst, are considered as other operational requirements. It is now possible to calculate the cost of the methanol production unit by dividing the total annual cost by the annual methanol production capacity, as:

$$C_F = \sum_i f_i C_{E,i} \quad (15)$$

where C_F , f_i , and C_E are the fixed capital cost for the complete system, the installation factor for equipment i , and the cost of equipment, respectively. The annual capital cost (CRF) and the annualized capital cost (C_A) of the plant is calculated by Eqs. (16) and (17), respectively [27]:

$$CRF = \frac{i(1+i)^n}{(1+i)^n - 1} \quad (16)$$

$$C_A = C_{CAP} \times CRF \quad (17)$$

where i is the discount rate, n is the lifetime of the project, and C_{CAP} is the initial capital cost. In this study, the discount rate was fixed as 18% to reflect the current economic status of Iran, and the lifetime of all equipment was assumed to be 10 years. The basic economic assumptions used in this study are specified in Table 1.

Profitability analysis of processes can be carried out by incorporating the payback period as an economic index [28]. The simple payback period of a plant can be estimated by dividing the total investment by the average annual cash flow [29]. In this approach, the assumption is that profit is generated forthwith and the whole capital is invested in the

Table 1
Economic assumptions for CO₂ hydrogenation.

Financial parameters	Values
Discount rate	18%
Lifetime of equipment	10 years
Annuity factor	0.22
CO ₂ cost	52 \$ Kg ⁻¹
H ₂ cost	6 \$ Kg ⁻¹
Methanol cost	350 \$ ton ⁻¹
Steam cost	4.4 \$ Kg ⁻¹
Cooling water cost	0.08 \$ Kg ⁻¹
Electricity cost	0.045 \$ Kw ⁻¹
Cost of methanol synthesis catalyst	131 \$ Kg ⁻¹
Cost of RWGS catalyst	20 \$ Kg ⁻¹

first year. This method of economic analysis assumes that all revenues begin immediately and that in year 0, all investments are made. The total investment is constituted from the fixed capital, working capital, and the variable costs of the plant. The total annual revenue from the selling of products is the average annual cash flow [30–32]

$$\text{Payback Period} = \frac{\text{Total Investment Cost}}{\text{Average Annual Cash Flow}} \quad (18)$$

4.1.3. Environmental analysis

The environmental analysis of the proposed process is investigated based on the net carbon dioxide reduction (NCR) for unit production of methanol [33]. Although CO₂ is consumed in the hydrogenation reactor, it is produced in the utility section. The difference between the total inlet and outlet flow of CO₂ is equal to the net CO₂ consumption. The CO₂ emission rate per unit of work consumption represents CO₂ generation. The total compression work, equivalent work of total heat output, the equivalent work of total heat input, and the work consumption of the air separation unit for oxygen feedstock production are the main sector of CO₂ emission [33].

$$\text{NCR} = \frac{\text{CO}_{2\text{CONSUMPTION}} - \text{CO}_{2\text{GENERATION}}}{\text{Methanol Rate}} \quad (19)$$

$$\text{CO}_{2\text{CONSUMPTION}} = \sum \text{CO}_{2\text{IN}} - \sum \text{CO}_{2\text{OUT}} \quad (20)$$

$$\text{CO}_{2\text{GENERATION}} = \text{EF} \left(\sum W_{Q\text{IN}} - \sum W_{Q\text{OUT}} + \sum W_{\text{COMP}} + W_{\text{ASU}} \right) \quad (21)$$

4.2. Process selection

Decision-making is the process of making choices by identifying a decision, gathering information, and assessing alternative resolutions. In psychology, decision-making is regarded as the cognitive process resulting in the selection of a belief or a course of action among several possible alternative options [34]. Every decision-making process produces a final choice, which may or may not prompt action [35]. In this research, TOPSIS as one of the multi-criteria decision analysis methods is applied to select the optimal operating condition of the process. It is based on the concept that the chosen alternative should have the shortest geometric distance from the positive ideal solution and the longest geometric distance from the negative ideal solution [36].

5. Results and discussion

5.1. Model validation

In this section, the simulation results of methanol synthesis reactor are validated against commercial methanol reactor. Table 2 shows the comparison between simulation results and plant data. The plant data was taken from a Domestic Chemical Company in Iran, in which synthesized gas as feedstock. It is observed that the plant data have a good agreement with the simulation data.

Table 2
Comparison between simulation results and plant data industrial reactor.

Item		Industrial reactor	Model	Absolute error %
Product mole Fraction	CO ₂	0.081	0.084	3.7
	CO	0.0238	0.0236	0.84
	H ₂	0.4897	0.4973	1.5
	CH ₄	0.2281	0.2264	0.74
	N ₂	0.0944	0.0936	0.85
	H ₂ O	0.0248	0.0224	9.6
	CH ₃ OH	0.0582	0.0542	6.9
Temperature (°C)		255	255	0
Pressure (kg cm ⁻²)		76.45	77.84	1.8
Flow (kmol hr ⁻¹)		6135.34	6181.36	0.75

5.2. Process synthesis and simulation

5.2.1. Direct conversion route

Fig. 2 presents the considered process for direct conversion of CO₂ to methanol. In the direct conversion route, the H₂ and CO₂ streams are compressed separately to 78 bar in a multi-stage compressor equipped with an interstage cooler. The required hydrogen could be supplied from sustainable processes including water electrolysis. After compression, the two gaseous streams are mixed and heated up to 230 °C in a heater. The mixture of CO₂ and H₂ at 78 bar and 230 °C feeds into the methanol synthesis reactor. The methanol synthesis reactor is a heat exchanger that syngas is converted to methanol in the tube side over CuO–ZnO/Al₂O₃ catalyst. The length, number of tubes and tube diameter of applied reactor are 7 m, 2996 tubes and 0.058 m. In the shell side, the saturated water flows to cool the tube side. The outlet stream from the methanol reactor contains methanol stream, and unreacted syngas. Thus, the non-condensable components are separated in a two-phase separator. The liquid stream leaving the flash drum, namely crude methanol is expanded to 2.5 bar and a temperature of 39 °C in the expansion drum and then sent to distillation columns. In the distillation section, water and methanol are separated. Fenske–Underwood–Gilliland Method has been used to design distillation column. After selecting light and heavy key components, the minimum number of equilibrium steps was determined from Fenske's relation. According to the calculations, the number of trays in the first column is 38 and the second one is 100. To determine the minimum reflux ratio, Underwood's relation is used and the values of it in the first column is 0.1 and in the second one 3.7 is considered. To determine the location of the feed tray, Fenske's relationship was used and the location of the feed tray was determined in the first column, tray 10, and in the next, tray 94 was determined. The outlet syngas from two phase separator is compressed and recycled to the methanol synthesis reactor after pressure and temperature regulation. To prevent the accumulation of methane in the methanol reactor, a part of the recycled stream is purged.

Based on the designed and simulated process, the overall conversion of carbon dioxide and hydrogen in the direct method are 88% and 85%. In addition, the of carbon dioxide conversion to the methanol in the methanol synthesis reactor is 28%. Considering an operation run time of 7800 h y⁻¹, about 354,197 tons of liquid methanol is produced annually. Fig. 3 show the molar fractions of H₂, CO₂, CO, H₂O, and methanol along the methanol synthesis reactor. It appears from Fig. 3 that H₂, CO₂ content decrease along the reactor while methanol and water molar fractions increases along the reactor. Since carbon monoxide synthesis is dominant over CO conversion to methanol at the first part of reactor, CO concentration approaches to maximum value of 2.8% and then decreases to 0.025.

Fig. 4 shows the temperature profile along the reactor. Since the net of reactions is exothermic, temperature increases and a hot spot is developed at the first half of reactor. Then, higher heat removal from reaction zone by the cooling medium compared to the generated heat through reactions, temperature decreases and approaches toward the coolant temperature at the last half of reactor.

5.2.2. Indirect conversion route

Fig. 5 presents the considered process for indirect conversion of CO₂ to methanol. The pressure of the gas stream is increased to 20 bar in a multi-stage compressor equipped with an interstage cooler. The feed steam feeds into the RWGS reactor to produce CO at 570 °C and 20 bar. The RWGS reactor is a adiabatic reactor to convert hydrogen and carbon dioxide to syngas over Ni/ Al₂O₃ catalyst. The length, and diameter of applied reactor are 4.5 m and 2.74 m. The outlet stream is cooled to 130 °C and feeds into a flash drum to separate water from the gasses mixture. After water removal, the pressure and temperature of the gas stream is increased to 78 bar and 230 °C and feeds to the methanol synthesis reactor, which consists of H₂, CO₂ and CO. The methanol reactor outlet is cooled to 40 °C, followed by condensation of methanol

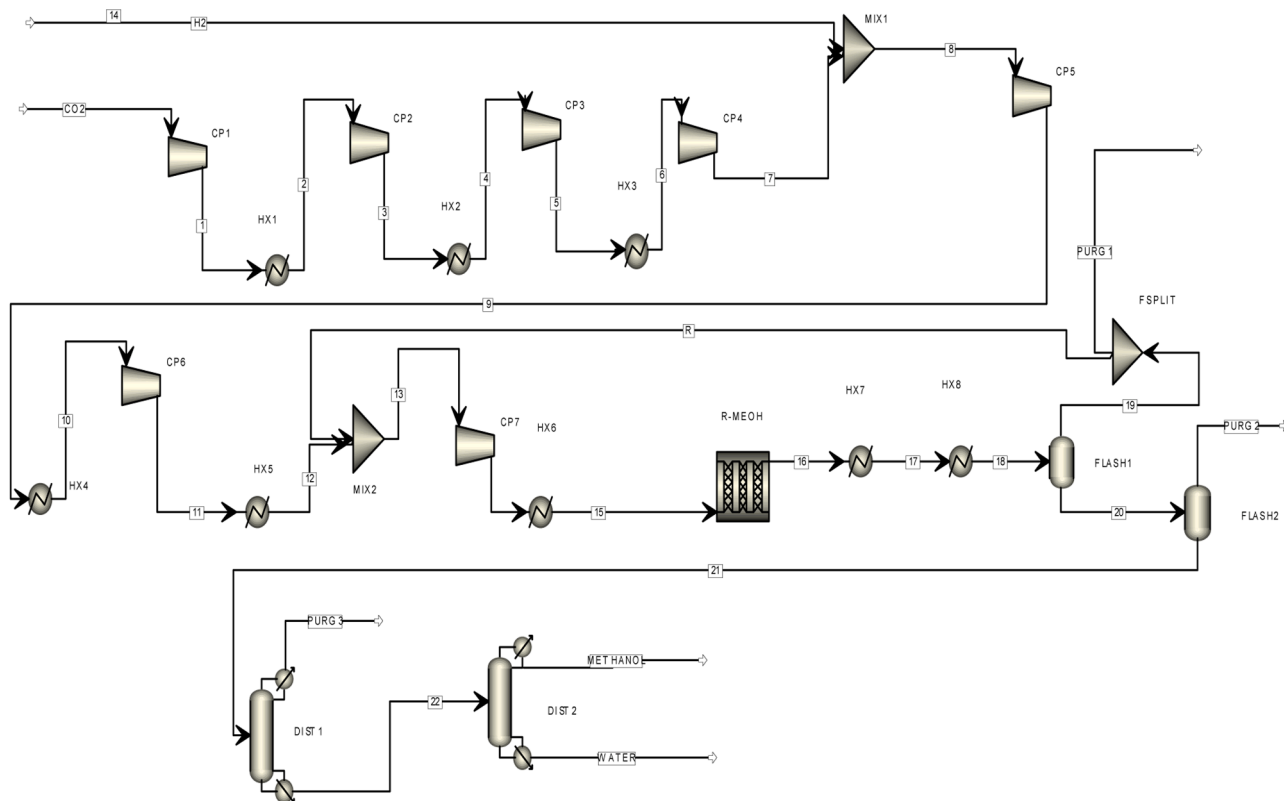


Fig. 2. The process flow diagram of direct conversion route.

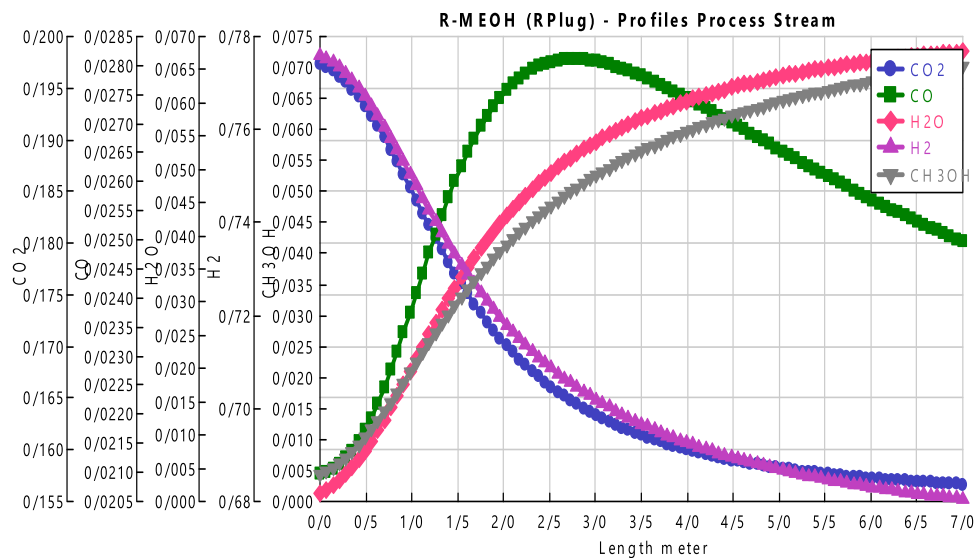


Fig. 3. Molar fraction of components along length of reactor in the direct conversion route.

and water in the applied separator. The mixture of water and methanol stream is sent to the distillation column in order to separate MeOH from water. Water leaves the column at the bottom, while MeOH leaves the column at the top as a liquid with a small amounts of water vapor. The outlet gas from separator is recycled to the methanol synthesis reactor.

In the indirect route, the reaction yield is about 26% and considering operation run time of 7800 h y^{-1} , the plant capacity is 362,237 tons of liquid methanol per annul. Based on the designed and simulated process, the overall conversion of carbon dioxide and hydrogen in the indirect method are 88% and 87%. Fig 6 (a, b) presents the mole fractions of H_2 , CO_2 , CO and H_2O along the RWGS and methanol synthesis reactors. In

the RWGS reactor, CO_2 and H_2 react and CO and H_2O produce. Based on the simulation results, mole fraction of CO and H_2O increases along the reactor and approaches to about 0.065. Fig. 6(b) shows the completely consume of H_2 , CO_2 , CO in the reactor and the decrement in the molar fractions along the reactor. Methanol and water molar fractions increases along the reactor and approaches about 0.085.

5.3. Sensitivity analysis

In order to check the best performance conditions of the proposed processes, the effect of change in feed flow rate and feed temperature

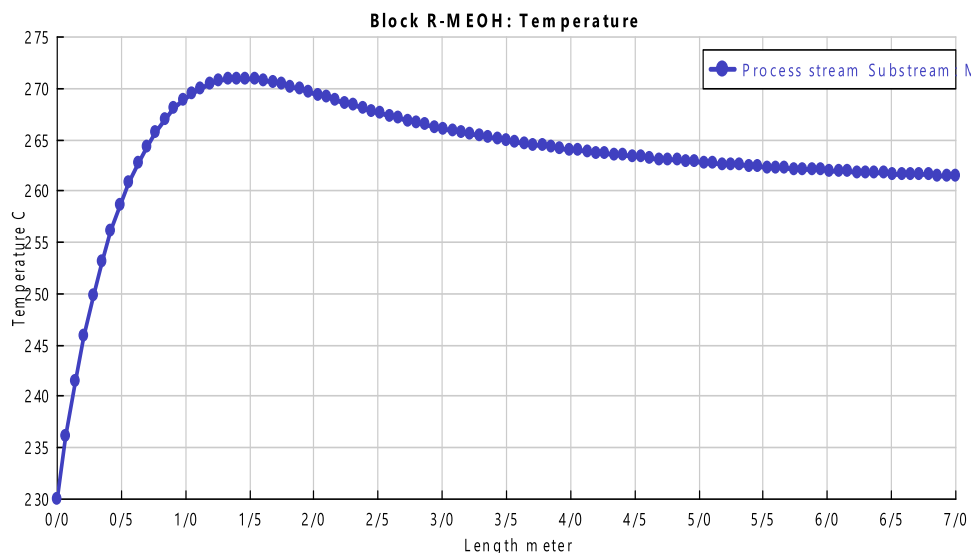


Fig. 4. Temperature profile along the methanol synthesis reactor in the direct conversion route.

and pressure conditions in the methanol reactor on the produced methanol have been investigated. Fig. 7(a, b) shows the effect of feed flow rate in the range 910 to 2049 kmol/hr for carbon dioxide and 2730 to 6147 kmol/hr for hydrogen on methanol production at a fixed H_2/CO_2 ratio of 3 in direct and indirect conversion routes.

The amount of methanol produced in the indirect method is 2.2% more than the direct conversion and this difference is shown in Fig. 7(a, b). In both processes, the amount of methanol produced is directly proportional to the increase of hydrogen. According to reactions (3), (4) and (5), with the increase of hydrogen in all three reactions, the amount of produced product increases. But the direct effect of carbon dioxide on the production of methanol can be seen only in reaction 5, which is also with a ratio of 1/3 compared to hydrogen.

Fig. 8 (a, b) shows the effect of the operating pressure in the range of 65 to 90 bar on the MeOH yield at a fixed H_2/CO_2 ratio of 3 and reaction temperatures, the range of 180 °C to 300 °C. It appears that higher operating pressures increased methanol yield. This effect was especially intensified in higher temperatures (>200 °C). Methanol production increases at higher pressures simply because, during CO_2 hydrogenation reactions, moles decrease in number. As a result, continued conversion of CO_2 to methanol at higher pressures is greater in the indirect route.

As reaction (4) (conversion of CO_2 to CO) is an endothermic reaction, its equilibrium is improved by increasing the temperature. On the contrary, reactions (3) and (5) (conversion of CO and CO_2 to methanol) are exothermic and an increase in temperature diminishes the equilibrium. It can, therefore, be stated that a lower temperature and a higher pressure increase the amount of methanol production.

As the methanol formation is an exothermic reaction, decreasing the temperature in the range of 300 to 180 °C has also a positive effect on the methanol yield and the flow rates of methanol and water. These findings can help in providing the optimal operating conditions to achieve higher efficiency in MeOH production.

5.4. Optimized reactor

In this section, the results of the optimized reactor are presented. It is mentioned that the selected decision variables are feed temperature, pressure, and coolant temperature. In addition, the methanol mass flow rate is selected as the objective function. Based on sensitivity analysis and studies results, coolant temperature is more effective compared to the feed temperature and has a considerable effect on the objective functions. The maximum methanol production rate is 1057.62 tons/day in direct route and 1070.28 tons/day for indirect route, respectively.

5.5. Economic analysis

The MeOH production cost was evaluated by an economic analysis. All direct and indirect costs of the process are calculated as a percentage of the fixed capital investment. Fig. 9 (a, b) shows the direct and indirect costs of both processes. Based on the applied analysis, the FCI of the direct and indirect processes are about 506 and 578 million, respectively. In addition, the net profit from the sale of methanol in the direct and indirect processes is about 132 and 135 million USD per year, respectively. Therefore, the rate of return on investment in the direct process is about 7 years, while in the indirect process is about 8 years.

For further investigation, the normalized FCI percentage are also tabulated in Table 3 for both direct and indirect costs, results of which are in the range reported by Peters et al. [24] and Smith et al. [25] showing the reliability of the economic analysis in this study.

5.6. Environmental analysis

Fig. 10 displays the results of the net amount of carbon dioxide reduction for unit production of methanol in the direct, indirect and conventional routes (CR). Before discussing the achievements of this figure, it should be considered that the total compression work is actually the power of the compressor to increase the stream flow pressure, the equivalent work of total heat output is related to the heat that is removed from the process through coolers, the equivalent work of total heat input is related to the heat that is logged into the process through heaters, the exact amount of which are obtained from the simulation results in Aspen Plus.

The emission factor is selected based on 3 tiers, each of which has its own conditions. While tier 1 depends on the carbon dioxide and fuel amounts, it does not influence by the operating conditions and its technologies. In tier 2, the amount of emission factor is expressed based on the laws of different countries, while in some burning method, technology, and operational conditions are important, in some others these parameters are not important. In tier 3, the amount of emission factor based on the laws of different countries, not only the amount of carbon dioxide, but also all types of pollution caused by burning, are calculated from detailed relationships.

Both conversion routes (direct and indirect) need carbon dioxide as a raw material, which is a source of CO_2 consumption. In conversion routes, there are approximately 1590.96 tons of CO_2 input daily. The total CO_2 output via direct and indirect routes is around 197.21 tons and 170.21 tons daily. Significant volumes of carbon dioxide are created

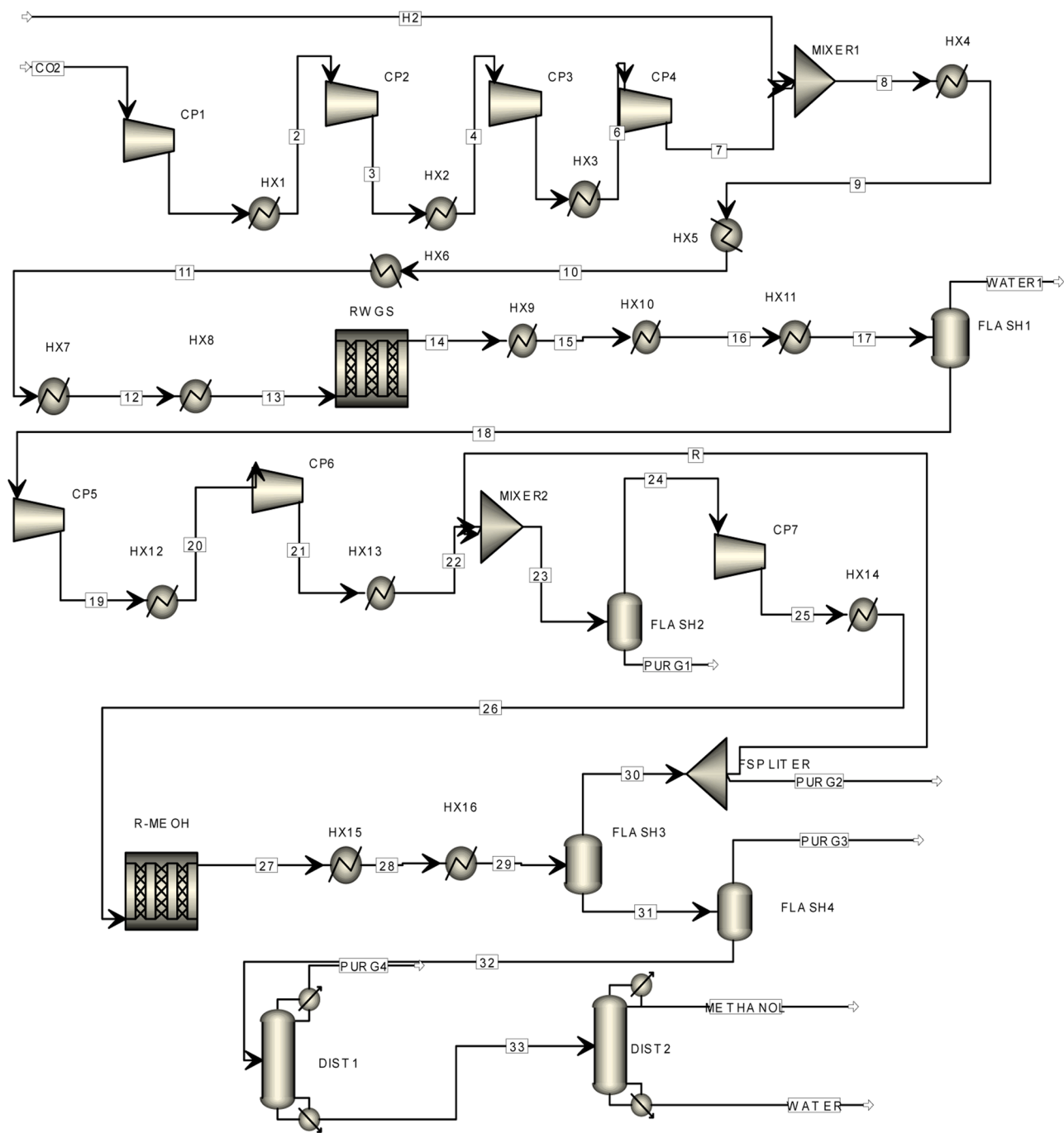


Fig. 5. The process flow diagram of indirect conversion route.

even though they are produced in the utility sector since the heat and electricity needed in the plant are produced by burning fossil fuels. In coolers and heaters, the input and output heat are therefore equal 1929.28, 6252 GJ/day in the direct route and 3426.72, 7097.76 GJ/day in the indirect route. The power of compressors is equal 1842.72 GJ/day in the first one and 1719.12 GJ/day in second one.

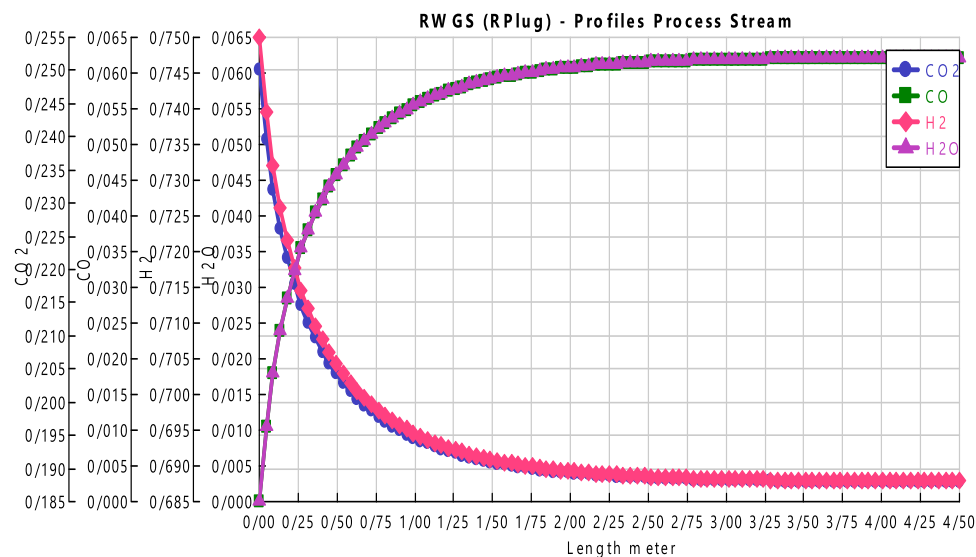
While in the conventional route, synthesis gas is used as feed, so the amount of carbon dioxide input is equal to 6387 tons/day and its output amount is equivalent to 6316 tons/day. In coolers and heaters, the input and output heat are 360 and 96 GJ/day, respectively and the power of compressors is equal to 528 GJ/day to produce 200 tons/day methanol.

When an industrial process consumes more carbon dioxide compared to released CO₂, it is considered as a green process and has a positive NCR number. This number indicates the amount of carbon consumption,

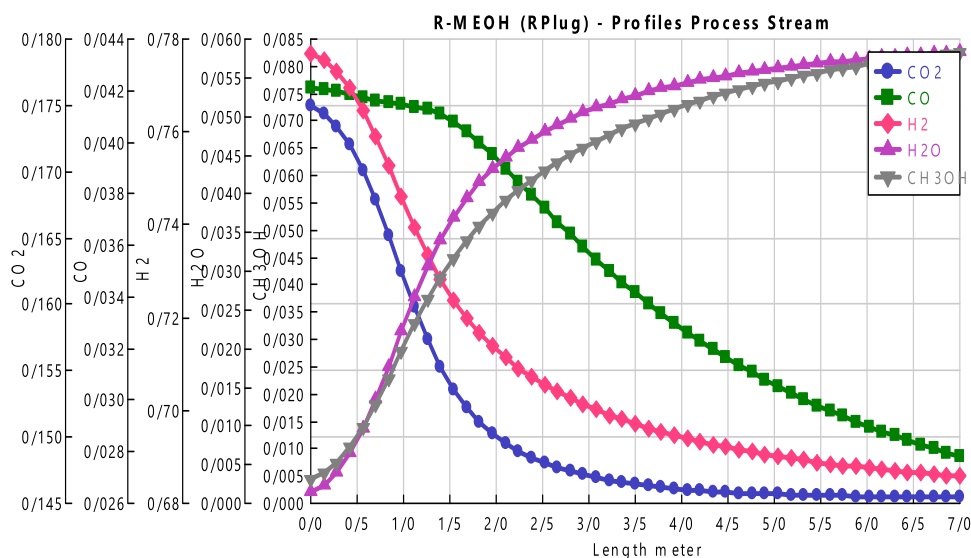
and the larger it is, the more environment-friendly the process. As displayed in Fig. 8, although, both direct and indirect processes consume CO₂, the direct method is more effective. Based on the simulation results the NCR of direct and indirect routes are 1.25 and 1.21, respectively.

5.7. Process selection

Although the direct conversion route presents a higher net amount of carbon dioxide reduction, lower fixed capital cost and lower return ratio is presents lower net profit compared to the indirect method. In this research, the best process is selected by TOPSIS as an efficient decision-making method. Table 4 presents the value of alternatives in direct and indirect processes. After normalizing and finding positive and negative ideals and scoring the criteria, the direct conversion route has been



(a)



(b)

Fig. 6. Molar fraction of reactants and products along (a) Methanol synthesis reactor (b) RWGS reactor.

selected as the most optimal method of methanol production from carbon dioxide hydrogenation.

6. Conclusion

The two processes for synthesizing methanol utilizing carbon dioxide as the starting material was contrasted in this research. Direct synthesis has been demonstrated to have superior economic and environmental assessment; however, the difference is largely inconsequential, particularly in the case of NCR. These technologies have the potential to reduce CO₂ emissions, therefore one might say that they are promising. In order to achieve adequate living conditions for our planet, greenhouse gas emissions must be reduced. Both technologies have drawbacks, but it is realistic to suppose that they might be fixed by new advances in the future. Numerous aspects should be taken into account when choosing which procedure would be appropriate for use in practice. First and foremost, the plant's location would be crucial. The two-step method may be more cost-effective, for instance, if the facility were

situated close to a source of abundantly inexpensive power. However, direct synthesis would be preferable if the factory were situated in an area subject to stringent environmental regulations since it would offer superior reactant consumption. Both procedures can be improved, and these changes may be made in the next years. By starting these modifications, the economic viability and energy efficiency might be improved while the investment costs may be reduced.

Additionally, the whole process—including CO₂ collection, water electrolysis, and methanol synthesis and separation—can be heated. Future research would look at the process's potential enhancements. With the aforementioned advances, it would be realistic to anticipate that one of these methods will make its way into practice.

From another point of view, using the direct route is preferable if the time to profit and the initial capital are crucial considerations. While the indirect technique is the best option if great process efficiency and output volumes are sought.

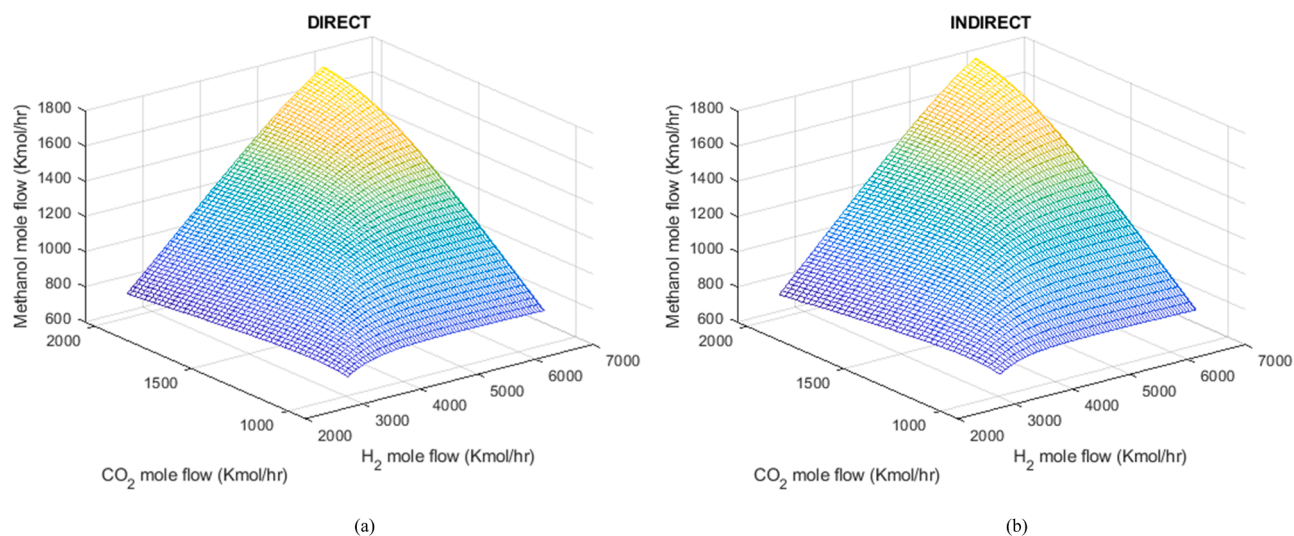


Fig. 7. Methanol production flow rate as a function of feed flow rate (a) direct (b) indirect.

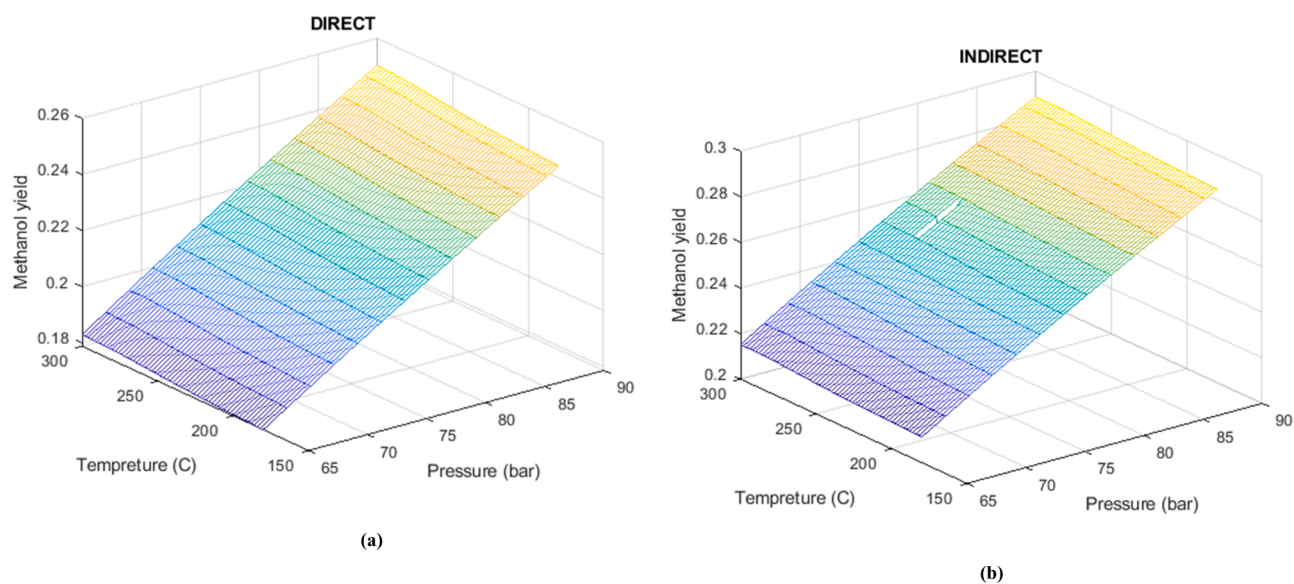


Fig. 8. Methanol yield as a function of inlet temperature and pressure in (a) direct and (b) indirect processes.

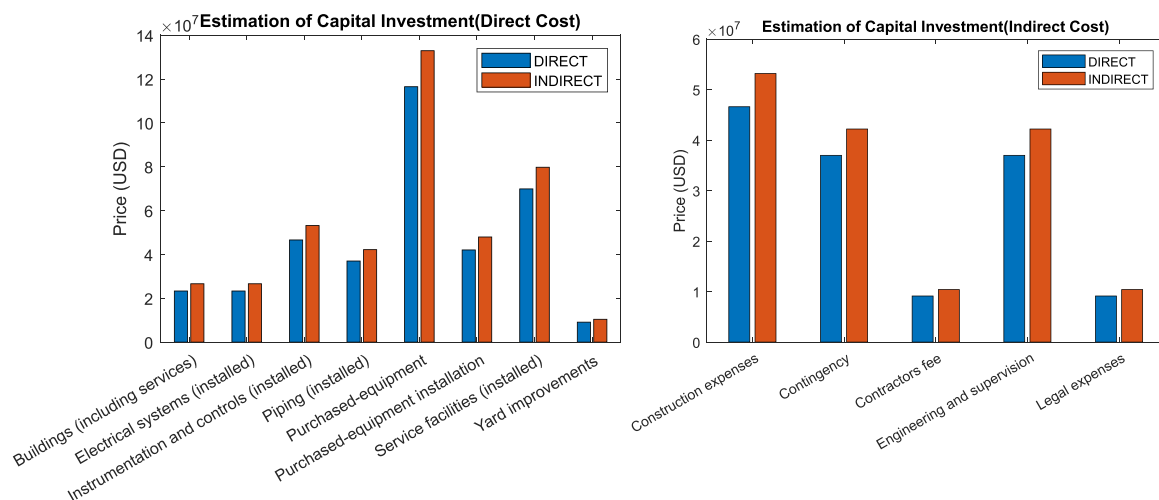


Fig. 9. Estimation of capital investment (a) direct cost (b) indirect cost.

Table 3
The estimation of FCI.

Component	Percentage of FCI [24,25]	Normalize percentage of FCI
Direct Cost		
Purchased-equipment	15–40%	23%
Purchased-equipment installation	6–14%	8.3%
Instrumentation and controls (installed)	2–12%	9.2%
Piping (installed)	4–17%	7.3%
Electrical systems (installed)	2–10%	4.6%
Buildings (including services)	2–18%	4.6%
Yard improvements	2–5%	1.8%
Service facilities (installed)	8–30%	13.8%
Indirect Cost		
Engineering and supervision	4–20%	7.3%
Construction expenses	4–17%	9.2%
Legal expenses	1–3%	1.8%
Contractor's fee	2–6%	1.8%
Contingency	5–15%	7.3%

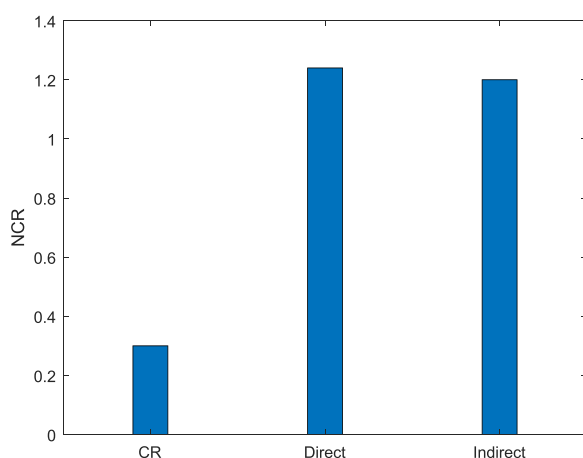


Fig. 10. NCR in direct and indirect processes.

Table 4
The value of alternatives in direct and indirect processes.

	NCR	FCI	Net Profit	ROI
Direct Conversion Route	1.24	506,000,000	132,000,000	7.4
Indirect Conversion Route	1.2	578,000,000	135,000,000	8.4

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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